Trigonometric identities

1 Express $\tan x$ in terms of $\cos x$ and $\sin x$.

2 Find the exact value of $\tan 10^\circ - \dfrac{\sin 10^\circ}{\cos 10^\circ}$.

3 Find the value of $\cos x$ given the following:

a $\sin x = -\dfrac{\sqrt{3}}{2}$ and $\tan x = -\sqrt{3}$

b $\sin x = \dfrac{b}{c}$ and $\tan x = \dfrac{b}{a}$

4 Find the value of $\sin x$ given the following:

a $\cos x = \dfrac{15}{17}$ and $\tan x = \dfrac{8}{15}$

b $\cos x = -\dfrac{5}{13}$ and $\tan x > 0$

5 Find the value of $\tan x$ given the following:

a $\sin x = \dfrac{4}{5}$ and $\cos x = \dfrac{3}{5}$

b $\sin x = \dfrac{2}{3}$ and $\cos x < 0$

6 Simplify the following expressions:

a $\tan\theta\cos\theta$

b $\sin(90^\circ - y)\tan y$

c $\dfrac{\sin\theta - \cos\theta}{\cos\theta}$

7 Find the exact value of $\tan x$ given the following equations:

a $\sin x = 2\cos x$

b $17\cos x - 31\sin x = 0$

c $6\sin x - 11\cos x = 0$

d $2\sin^2 x - \cos^2 x = 0$

Pythagorean identities

8 State whether the following statements are correct:

a $\sin^2\theta + \cos^2\theta = 1$

b $\sin^2\theta + \cos^2\theta = 2$

9 Find the exact value of the following:

a $\sin^2(20^\circ) + \cos^2(20^\circ)$

b $4\sin^2(20^\circ) + 4\cos^2(20^\circ)$

10 Simplify the following expressions:

a $(\cos\theta - 1)(\cos\theta + 1)$

b $(\cos\theta - \sin\theta)^2$

c $\cos\theta \sin^2(\theta) - \cos\theta$

d $(3 - \cos x)^2 + \sin^2(x)$

e $\dfrac{\sin^2(\theta)}{1 - \sin^2(\theta)}$

f $\dfrac{1 - \sin^2(\theta)}{\sin^2(\theta) + \cos^2(\theta)}$

11 If $x = 4\sin\theta$ and $y = 3\cos\theta$, form an equation relating x and y that does not involve $\sin\theta$ or $\cos\theta$.

Proofs

12 Prove the following identities:

a $\dfrac{\cos x \tan x}{\sin x} = 1$

b $\dfrac{1 - \sin^2(x)}{\cos x} = \cos x$

c $\dfrac{\sin\theta}{1 - \cos\theta} = \dfrac{1 + \cos\theta}{\sin\theta}$

d $(\sin x + \cos x)^2 = 1 + 2\sin x \cos x$

e $\sin A \cos A \tan A = \sin^2(A)$

f $\cos^4(x) - \sin^4(x) = 2\cos^2(x) - 1$

g $5\cos^2(\theta) - 3 = 2 - 5\sin^2(\theta)$

h $\dfrac{(1 + \sin\theta)^2 + \cos^2(\theta)}{1 + \sin\theta} = 2$

i $\dfrac{\sin^2(x) + \sin x \cos x}{\cos^2(x) + \sin x \cos x} = \tan x$

j $\dfrac{\sin x \cos(90^\circ - x)}{\cos x \sin(90^\circ - x)} = \tan^2(x)$

13 Prove the following identities:

a $\tan^2(y) - \sin^2(y) = \tan^2(y)\sin^2(y)$

b $\sin^2(a) - \sin^2(b) + \cos^2(a)\sin^2(b) - \sin^2(a)\cos^2(b) = 0$

c $\left(1 - \sin^2(x)\right)\left(1 + \sin^2(x)\right) = 2\cos^2(x) - \cos^4(x)$